EEG-based discrimination of different cognitive workload levels from mental arithmetic

Zheng Yang Chin, Xin Zhang, Chuanchu Wang, Kai Keng Ang

Abstract— **Cognitive workload, which is the level of mental effort required for a cognitive task, can be assessed by monitoring the changes in neurophysiological measures such as electroencephalogram (EEG). This study investigates the performance of an EEG-based Brain-Computer Interface (BCI) to discriminate different difficulty levels in performing a mental arithmetic task. EEG data from 10 subjects were collected while performing mental addition with 3 difficulty levels (easy, medium and hard). EEG features were then extracted using band power and Common Spatial Pattern features and subsequently features were selected using Fisher Ratio to train a Linear Discriminant Classifier. The results from 10-fold cross-validation yielded averaged accuracy of 90% for 2 classes (easy versus hard tasks) and 66% for 3 classes (easy versus medium versus hard tasks). Hence the results showed the feasibility of using EEG-based BCI to measure cognitive workload in performing mental arithmetic.**

I. Introduction

Cognitive workload or mental workload refers to the level of mental effort required to perform a task such as problem-solving, reading and writing, and performing mental arithmetic. Heavy cognitive workload for a prolonged period of time could lead to cognitive overload. This results in issues such as a drop in learning gain [1], affecting task performance [2]or even critical errors in situations where high levels of sustained attention are required. In contrast, low cognitive workload could lead to cognitive underload, as the task comes too easy, resulting in boredom and reducing the learning gain as well.

Hence identification of mental workload during cognitive tasks is an important area in cognitive neuroscience and other real-world applications. The development of a real-time assessment of cognitive workload could lead to building smart systems to adapt training content and keep subjects in an optimal range of cognitive workload to ensure learning progress[1] or to even proactively warn subjects about their stress during these workload states [3].

The level of cognitive workload depends on factors such as prior experience, physical condition, and varies from person to person. Hence, there are conventional methods to measure cognitive workload, which include verbal or written feedback from the subject's point of view. For example the NASA-Task Load Index (TLX) subjective ratings could be used to evaluate cognitive workload of system operators. However, the assumption is that the subject could accurately identify and report changes in cognitive workload, which may be subjected to confirmation biases [4]. Furthermore, to evaluate the cognitive workload of many subjects concurrently based on subjective feedback in real time e.g. 30-40 students in a classroom for a teacher would be challenging [2].

To complement conventional methods, physiological measures such as Galvanic Skin Response (GSR), heart rate, as well as brain signals, have been investigated for their effectiveness to provide real-time assessment or classification of cognitive workload. Existing studies have investigated the changes in cognitive workload during mental arithmetic. In previous work [5],[6], the authors measured Near-Infrared Spectroscopy (NIRS) brain signal changes when subjects were performing a mental arithmetic task involving digit additions. The task was categorized into 2 difficulty levels which varied in the number of digits involved and the presence / absence of a carry-over operation. They proposed classification algorithms which yielded an overall intra-subject accuracy of 73% in classifying hard versus easy mental arithmetic tasks and 92% in classifying hard task versus a neutral state.

Besides NIRS, changes in scalp-recorded brain signals or electroencephalogram (EEG) is also employed to detect changes in mental workload [7]. Due to its relatively higher temporal resolution, EEG can capture fast and dynamically changing brainwave patterns in complex cognitive tasks. Examples of studies that employ EEG-based classification of mental arithmetic are as follows: In [8], the authors trained their classifier to discriminate between 3 conditions: relaxed, low workload and high workload using mental arithmetic, which obtained an average accuracy of 62%. In [9], the authors proposed a generalized Higuchi fractal dimension spectrum method for mental arithmetic task recognition, which yielded a subject-dependent accuracy of 97.9% for mental arithmetic versus the relaxed state. In [1], the authors proposed an inter-subject classifier to classify EEG during 3 levels of mental arithmetic task difficulty. Their classifier yielded around 56% accuracy averaged over 10 subjects, which could classify easy and medium tasks at 68% and 84% but below chance level at 16% for hard tasks. However, one limitation that the authors identified was the tasks were presented in order of increasing difficulty level and hence classification accuracy could have been affected by

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increasing levels of fatigue by the time the subjects perform the hard tasks.

As an extension of the previous work carried out on NIRS [5],[6], this paper investigates the use of EEG with signal processing and machine learning algorithms to differentiate difficulty levels of a mental arithmetic task. Developing methods that could classify different levels of cognitive workload that arises due to different mental arithmetic tasks is a step towards complementing existing strategies to get the real-time feedback on the subjects' cognitive workload in numerical learning. Hence, this could potentially lead to adaptation of training programs and learning environments to the subjects' individual prerequisites [1]

II. METHODS

A. Mental Arithmetic Tasks

Subjects were required to solve mental arithmetic tasks where they performed mental addition of two numbers. The difficulty levels were defined based on the Q-value measure proposed in [1],[10], which takes into account both the problem size and the presence of the carry-over operation. Mental arithmetic tasks with more digits and with carry over operations will have a higher Q-value. Tasks were divided into 3 classes as easy (QE), medium (QM) and hard (QH) and examples of such tasks have been summarized in Table 1. For details of the Q-value, the reader is invited to refer to [1] for more details.

B. Data Acquisition

EEG data were collected using a 40-channel Neuroscan EEG cap, with a sampling frequency of 250Hz, from 10 healthy subjects. The setup is shown in Figure 1. The experimental procedures involving human subjects described in this paper were approved by the Institutional Review Board. Each subject went through 4 rounds of the experiment. Each round consisted of 9 blocks, where each block presented 5 trials of the same difficulty level. The order of the blocks was randomized. In each trial, the mental arithmetic task is presented. In total, there were 4 x 9 x 5 = 180 trials, equally distributed among the 3 difficulty levels. The protocol for each trial is shown in Figure 2. The subject was instructed to minimize movement when the task was presented for 8s.

Table 1: Mental Arithmetic task difficulty categorized according to Q-values and examples provided. The Q value accounts for the number of digits and the presence of the carry over operation

Task Difficulty	QE (Easy)	QM(Medium)	QH(Hard)
Q-value	$Q < 2$	$2 \leq Q < 4$	$Q \geq 4$
No. of tasks in this study	60	60	60
Example	3+5 Q = log[3+5+(3+5)] = 1.2	47 + 46 Q = log [4+4+8 +10+(8+10)]+ log[7+6+13+1+ (13+1)] = 3.0	791 + 349 Q = 4.7

C. Data Extraction and Inspection of EEG Artifacts

The continuous EEG data from the 30 channels F7, F3, Fz, F4, F8, FT7, FC3, FCz, FC4, FT8, T7, C3, Cz, C4, T8, TP7, CP3, CPz, CP4, TP8, P7, P3, Pz, P4, P8, O1, Oz, O2, PO1, PO2 were then segmented into 180 trials. Each trial consisted of EEG during the time segment of [0s, Ts], $T \leq 8$, relative to the appearance of each mental arithmetic task. The trials were then visually inspected in EEGlab and trials with high amplitude artifacts or noise were removed from subsequent processing. ~ 4.4 trials were removed per subject.

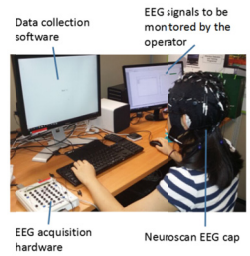

Figure 1: setup for the subject performing mental arithmetic

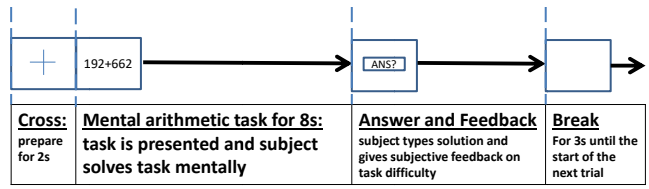

Figure 2: protocol for each trial. In each trial, a task is presented for 8s where the subject needs to solve a mental arithmetic task. The subject then types in his answer and gives subjective feedback on the task difficulty (easy/medium/hard). An inter-trial break of 3s is given until the next trial.

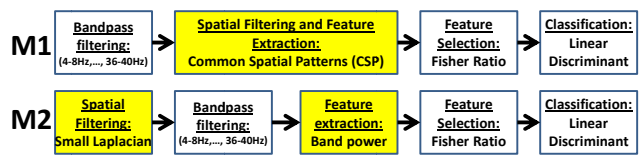

Figure 3: Methods M1 and M2 are employed to process and classify the EEG during mental arithmetic tasks. The main difference is M1 carries out spatial filtering and feature extraction using the CSP algorithm while M2 employs the Small Laplacian spatial filter with EEG band power feature extraction. In both methods, temporal filtering is carried out using 9 bandpass filters (4 – 40Hz), feature selection and classification is carried out using Fisher Ratio and and Linear Discriminant Analysis (LDA)

D. EEG Processing Methods

Two methods M1 and M2 are used to process the EEG. Both methods are summarized in Figure 3. In both methods, temporal filtering is carried out using 9 Chebyshev Type II bandpass filters which splits the EEG signal into 9 bands from 4-8Hz, 8-12Hz,…, 36-40Hz. The main differences between M1 and M2 are:

- M1 employs the Common Spatial Pattern (CSP) algorithm [11] to perform spatial filtering and CSP features are extracted
- M2 employs the small Laplacian spatial filters[12] to perform spatial filtering and band power features are extracted

E. Feature Extraction using CSP / Band power and Feature Selection using Fisher Ratio

In M1, only the top 2 and bottom 2 CSP features were extracted [11],[13]. The CSP feature extraction is based on maximizing the difference between 2 classes using a simultaneous diagonalization approach. Hence the number of CSP features differ based on the number of classes. For a 2-

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class problem, the number of features extracted for each trial was 9 bands x 4 features = 36 features. For a 3-class problem, a one-versus-rest approach [13] to CSP feature extraction was taken, and the number of features extracted for each trial was 9 bands x 4 features x 3 classes = 108 features. In M2, band power features were extracted from all 30 channels, hence there were 9 x 30 = 270 features for each trial.

Fisher Ratio (FR) feature selection is then subsequently employed to select a set of the most discriminative features for classification. The FR for each feature is defined as the ratio of the variance of the between classes S_B to the variance of the within classes S_w, shown in (1),(2) and (3).

$$\text{FR} = s_B / s_w \tag{1}$$

$$s_B = \sum_{k=1}^{C} \frac{n_k}{n} \left(m_k - m \right)^2 \tag{2}$$

$$s_w = \sum_{k=1}^{C} \frac{n_k}{n} \sum_{j=1}^{n_k} \left(x_{kj} - m_k \right)^2 \tag{3}$$

where C is the total number of classes, n_k is the number of trials in class k, n is the total number of trials, m_k is the mean of the feature for class k, m is the overall mean of the feature, x_{kj} is the feature for the j-th trial in class k. The Fisher Ratio of each feature was computed and 16 features with the highest Fisher Ratio were selected for classification.

F. Classification

The Linear Discriminant Classifier [14] in (1) was employed, where the discrimination rule is to assign the trial $\mathbf{x}$ to class c if $g_c \geq g_k$ for all $k \neq c$, $k = 1, \ldots, C$.

$$g_k = \mathbf{x}^T \Sigma^{-1} \boldsymbol{\mu}_k + \left(\log p_k - 0.5 \boldsymbol{\mu}_k^T \Sigma^{-1} \boldsymbol{\mu}_k \right) \tag{4}$$

where $\mathbf{x}$ is the feature vector for a trial, $\sum$ is the pooled covariance matrix, $\boldsymbol{\mu}_k$ is the mean feature vector for Class k, $p_k = n_k / n$ is an estimate of the prior probability of each class.

III. EXPERIMENT RESULTS AND DISCUSSION

A. Subject's mental arithmetic performance and subjective feedback on task difficulty

Table 2 shows the subjects' mental arithmetic performance in providing the correct solution for each difficulty level. As the difficulty level of the tasks increases, the number of incorrect answers also increases. The subjects also provided their subjective feedback on the difficulty level of the 180 tasks: and the subjective ratings for 2 subjects are also shown in Table 3. Some subjects e.g. S6 gave subjective ratings which were similar to the Q-value ratings. However other subjects e.g. S7 perceived the tasks to be easier, where all the 60 hard tasks were rated as medium (SM) and easy difficulty (SE).

B. Intra-Subject Classification Performance

The 10-fold cross validation (CV) accuracies averaged over all subjects for the trial time segments of 1s to 8s are presented in Figure 4 for 3-class (easy versus medium versus hard i.e QE v QM v QH) and 2-class classification for all pair-wise combinations (QE v QH, QE v QM and QM v QH). The accuracy increases as the time segment increases from 1s to 8s. For example, M1 which computes CSP features yielded around 90% accuracy around 5s onwards for the easy versus hard task. One possible reason why this occurs is because the hard task e.g. 791 + 349 takes a relatively longer time to solve as compared to the easy task e.g. 3 + 5.

Table 2: accuracy of the 10 subjects' solutions to the mental arithmetic tasks, according to the difficulty of the tasks rated using Q-Value (QE = Easy, QM = Medium, QH = Hard). The number of errors increases as difficulty level increases.

Task type	**Subject's accuracy in solving the mental arithmetic tasks (%)**									
	1	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***
QE	98	98	98	98	100	100	100	100	100	100
QM	93	100	100	95	98	93	95	100	95	98
QH	90	77	93	87	97	70	82	92	77	92

Table 3: Comparison of task difficulty rated by Q-value (QE, QM and QH) versus subjective ratings (SE, SM and SH) for some subjects. E, M and H represents Easy, Medium and Hard respectively. Subjective ratings provided by S6 are similar to the Q-value ratings. However S7 gave an easier subjective rating for the more difficult tasks (QM as SE and QH as SM)

Task Type	**Subject's perceived rating of task difficulty**					
	S6			**S7**		
	SE	SM	SH	SE	SM	SH
QE	60	0	0	60	0	0
QM	3	47	10	53	7	0
QH	0	5	55	3	57	0

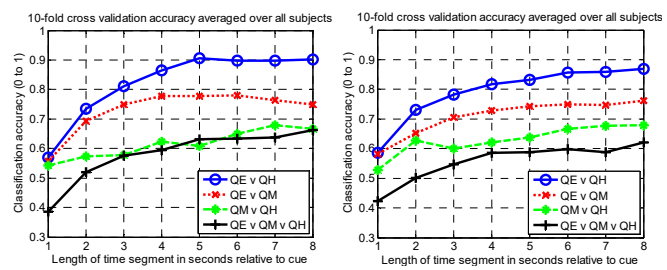

Figure 4: 10-fold CV accuracy as the time segment of each trial increases from 1 to 8s after the onset of the task cue. (Left) uses CSP features; (Right) using Band power features.

For 2-class classification, M1 and M2 yielded the highest accuracies for easy versus hard tasks, compared to easy versus medium tasks, and medium versus hard tasks. For 3-class classification, M1 and M2 yielded ~60% accuracy (33% naïve) around trial time segments of 5s and longer. For the 8s time segment, M1 yielded relatively the highest accuracy of 66.1%, which meant that the 3-class classifier could discriminate 2 out of the 3 classes on average. One possible reason for this could be because some subjects did not feel there was a difference between consecutive difficulty levels as shown in Table 3. Hence future studies could also examine the effect of including subjective feedback as part of the classification process to improve the sensitivity of discriminating multiple levels of cognitive workload.

Table 4: confusion matrices for M1 and M2. The easy and hard tasks are discriminated relatively better than the medium tasks.

		M1 (CSP)			***M2 (Band power)***		
		Predicted Class			***Predicted Class***		
		QE	QM	QH	QE	QM	QH
True Class	QE	73%	21%	6%	72%	22%	6%
	QM	16%	58%	26%	22%	49%	30%
	QH	4%	29%	67%	6%	28%	66%

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Table 4 shows the breakdown in the classification of the 3 classes, as confusion matrices for M1 and M2. In both cases, the easy and hard tasks can be discriminated much better than the medium tasks. The difference in performance in M1 and M2 could be attributed to an improved discrimination of the medium task (58%) versus M2 (49%).

C. Comparison with existing NIRS BCI System for Mental Arithmetic EEG Recognition

The previous high density NIRS study [5] used a different definition of easy and hard tasks, where easy tasks were adding 3-digits number with 2-digits number without carry over, and hard tasks were adding 4 digits with 3 digits with at least 1 carry over. The authors' approach yielded around 72.6% 5 x 5 fold CV accuracies for easy versus hard tasks averaged over 8 subjects. Hence, the results in this study are comparable to those in the previous study. Furthermore the EEG setup used a smaller number of channels (30) compared to the NIRS setup, which required 1024 channels

D. Analysis of features selected

Figure 5 shows the distribution of the top 3 selected discriminative CSP (M1; left) / band power features (M2; right) for easy versus hard tasks according to the frequency bands. In both cases, the top 3 features were mostly computed from the 4-16Hz, which encompass theta, alpha and lower beta bands. This supports findings from other studies, which have suggested theta and alpha band oscillations play an important role during the execution of mental arithmetic [1],[15]. For some subjects e.g. S3, selected features were also computed from frequency band 7 (28-32Hz) around the higher beta and gamma bands, which other studies[16] have also identified changes in the gamma band in the parietal lobe during cognitive task processing.

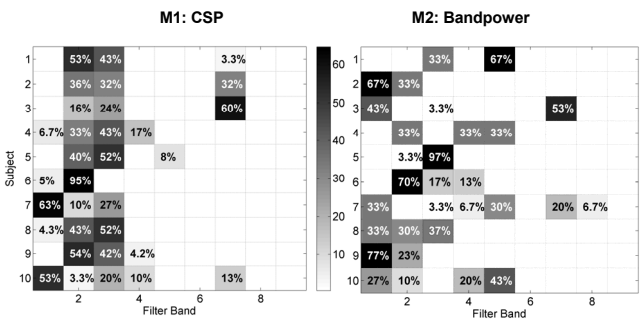

Figure 5: Distribution of the selected frequency bands (x-axis) for each subject (y-axis), based on top 3 selected features (M1, M2) to classify Easy versus Hard tasks and 8s EEG time segment. There are 9 frequency bands (4-8Hz, 8-12Hz,…36-40Hz). Most of the top 3 features were from Bands 2 and 3 which encompasses 4-12Hz (theta band and alpha band)

IV. CONCLUSION

This paper investigated EEG-based classification of cognitive workload by discriminating 3 different levels of mental arithmetic tasks (easy, medium and hard difficulties). EEG Band power features and Common Spatial Patterns (CSP) features were extracted. Subsequent feature selection using Fisher Ratio showed that the best discriminating features were computed from the theta and alpha bands, which are associated with cognitive task processing. Classification using the LDA classifier, yielded a 10-fold cross validation accuracy of about 90% for easy versus hard tasks and 66% for easy versus medium versus hard tasks. The easy versus hard tasks could be discriminated relatively the best amongst all possible pairs of the 3 tasks. A longer time segment ≥ 5s improves the classification accuracy.

Hence, the classification accuracies could be influenced by these factors: the difference in the task nature and the length of EEG time segment since the subject would require more time to solve hard tasks. In summary, the results in this study provide a starting point in the development of a real-time EEG-based monitoring system, to complement existing subjective measures of the subject's cognitive workload during numerical learning and other cognitive tasks.

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